UMAIR MANZOOR

Software Engineer — Systems & Backend

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Systems & Backend Engineer with 2+ years building production systems in C# and C++20, real-time applications, client-server architecture, and low-latency performance optimization. Shipping commercial releases and engineering sub-microsecond matching engines, custom memory allocators, and modular runtime engines from scratch.

PROFESSIONAL EXPERIENCE

Software Engineer 01/2026 – Present

ZnK, GenIteam Lahore, Pakistan

- Architected and integrated AWS cloud infrastructure (S3, CloudFront) for short video media streaming and telemetry across live services serving **10M+ combined downloads**.
- Profiled and eliminated runtime heap allocations and GC churn in core C# execution loops, stabilizing frame delivery at a constant **60 FPS** on resource-constrained Android devices.
- Deployed automated telemetry pipelines and client-server synchronization protocols, maintaining uninterrupted service availability and reducing runtime error rates during multi-million user live-ops events.

Jr. Software Engineer (remote) 01/2025 – 12/2025

EnsembleUI New York, USA

- Optimized core declarative client runtime pipelines in Flutter/Dart, reducing draw-call overhead via fine-grained render caching and asynchronous backend-to-widget stream loading.
- Authored modular reference architectures adopted by **100+ external mobile developers**; established standardized client-server state synchronization patterns.
- Maintained modular package architecture across authentication, networking, and local storage within a Melos monorepo, accelerating multi-platform build and integration cycles.

Software Engineer Intern 09/2024 – 12/2024

UET Game Studio Lahore, Pakistan

- Engineered low-latency interaction and physics synchronization systems for medical simulation software operating under strict mobile hardware thermal and memory budgets.
- Optimized GPU draw call batching and vertex memory layouts, cutting draw calls by **35%** and shipping 2 production applications with **10K+ downloads** on Google Play and Amazon Store.

PROJECTS

Otterus2D — Modular C++20 / OpenGL 4.5 Runtime Engine 05/2026 – Present

C++20, OpenGL 4.5 Core, Dynamic Batching, Decoupled Static Libs, Lua 5.3, EnTT

- Architected a modular systems runtime and batch renderer from scratch in C++20 and OpenGL 4.5 Core, batching textured quads into shared dynamic vertex buffers to minimize GPU draw calls.
- Organized engine into isolated static libraries with explicit interface boundaries (Renderer, Math, ECS, Audio); embedded Lua 5.3 via Sol3 for runtime script execution without recompilation.

Ensemble — Declarative Cross-Platform Client Runtime 2025

Flutter / Dart, Declarative AST Evaluation, Reactive State, Melos Monorepo

- Maintained client runtime subsystems that parse structured declarative YAML configurations into native widget hierarchies with dynamic state synchronization and render caching.
- Introduced new features, widgets, authentication methods and integrated accessibility requirements for apps built with platform.

Holt — Low-Latency HFT Electronic Matching Engine 01/2026 – Present

C++20, Lock-Free SPSC, Memory Pool, Monotonic PMR, Linux

- Engineered a deterministic HFT matching engine achieving **1.48M matches/sec** with sub-**420ns p99.9 latency**, compiled with -fno-exceptions and CPU core-pinning.
- Implemented wait-free SPSC circular ring buffers with **alignas(128)** padding between atomic cursors; structured price levels with monotonic PMR arena pools for **zero heap allocations**.

Custom Memory Allocator Suite 2026

C++20, Linear Bump Arena, LIFO Stack, Intrusive Free-List Pool, Doctest, Raylib

- Built a deterministic memory management suite featuring Linear Bump Arenas (**O(1)** allocation), LIFO Stacks with Marker snapshots, and Intrusive Free-List Pools with 0% external fragmentation.
- Engineered an STL-compliant TrackingAllocator wrapper and validated zero-leak invariants across **100% passing Doctest unit tests** and an interactive Raylib visualizer.

Archetype Entity Component System Runtime 2026

C++20, Data-Oriented Design, Type Erasure, Variadic Templates

- Handcrafted a data-oriented archetype ECS storing components in dense parallel contiguous column arrays, saturating CPU L1/L2 prefetchers and eliminating pointer indirection.
- Achieved **O(1)** entity deletions via swap-and-pop buffer relocation, and designed variadic template query iterators utilizing compile-time component IDs and bitset filtering.

Little King Unchained — Commercial Systems Architecture 2026

C#, State Machines, Execution Loops, GC Elimination, Steam PC Release

- Served as Main Code Engineer for commercial release on Steam; engineered decoupled hierarchical state machines, fixed-timestep execution loops, and asynchronous input abstraction layers.
- Profiled and eliminated runtime heap allocations to prevent garbage collector hitches, maintaining consistent 60 FPS delivery across varied PC hardware.

EDUCATION

BS Computer Science 10/2020 – 10/2024

University of Engineering & Technology, Lahore Lahore, Pakistan

Relevant Coursework: Operating Systems, Computer Architecture, Distributed Systems, Data Structures & Algorithms, Computer Networks, Database Systems, Software Design & Architecture.

SKILLS

Languages: C++, C#, Dart, Java, Lua, C, SQL

Systems & Architecture: Data-Oriented Design (ECS), Real-Time Systems, Object-Oriented Design, Memory Allocators & Pooling, Performance Profiling

Backend & Tools: Git, CMake, Unity, AWS, Visual Studio, Raylib, OpenGL